

GPTs are GPTs

An Early Look at the Labor Market Impact Potential of Large Language Models

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Reading Group Discussion · Presented by Bohan Bai

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Roadmap

- 1 Motivation & Context
- 2 Prior Literature & Framework
- 3 What Is a “GPT” (Both Meanings)
- 4 Methodology
- 5 Results
- 6 The GPT Argument (Second Meaning)
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- 8 Connections & Follow-Ups
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Terminology cheat sheet

Term	What it means
LLM / GPT (model)	Large Language Model. Generative Pre-trained Transformer. Examples: ChatGPT, GPT-4, Claude.
GPT (economic sense)	General Purpose Technology. A foundational technology that reshapes the whole economy. Historical examples: steam, electricity, the computer. (Bresnahan & Trajtenberg 1995)
Task-based framework	Autor's approach: jobs are bundles of tasks. Technology substitutes for some tasks and complements others.
O*NET DWA	Detailed Work Activity. Small, standardized task unit from the US Department of Labor's occupational database. About 19,000 DWAs across 900 occupations.
Exposure (E0/E1/E2)	rubric The paper's rating for each task. E0: LLM cannot accelerate. E1: LLM alone accelerates by 50%+. E2: LLM plus supporting software accelerates by 50%+.
Employment polarization	Empirical pattern where middle-wage jobs collapse while top and bottom grow. Documented in US, UK, Germany over 1980–2020.
Displacement effect	Technology moves tasks from labor to capital, reducing labor share (Acemoglu & Restrepo)

Why this paper matters

- ChatGPT launched Nov 2022; GPT-4 released March 2023.
- Immediate policy question: **how much of the workforce will be affected?**
- Before this paper: mostly speculation, hype, and Frey–Osborne-style estimates (2013) built on pre-LLM assumptions.
- After this paper: a defensible, task-level methodology that anchored virtually every downstream policy discussion.
- Published as arXiv preprint March 2023; later in *Science* 2024.

The paper's core question

Which tasks, across which occupations, could large language models substantially accelerate, and how is that exposure distributed across wages, skills, and industries?

The two questions in one title

1. Are **GPTs** (generative pre-trained transformers) capable of doing much of the work humans currently do?
2. Do **GPTs** (in the economic sense: *General Purpose Technologies*) qualify as such, like steam, electricity, or the computer?

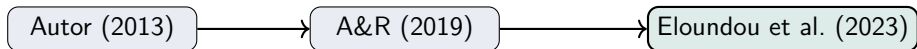
Preview of the answer

Yes to both. The exposure is broad ($\sim 80\%$ of workers), skewed toward *higher*-wage cognitive work (a reversal of historical automation), and the paper argues LLMs meet the classical criteria for a GPT: pervasive, improving, innovation-spawning.

Where this paper stands in the literature

Three streams it builds on

- 1 **Task-based framework:** Autor, Levy, Murnane (2003); Autor (2013)
Work = bundle of tasks; tech substitutes for some, complements others.
- 2 **Displacement / reinstatement:** Acemoglu & Restrepo (2019)
Tech displaces old tasks AND creates new ones; net effect depends on the balance.
- 3 **Occupational exposure measurement:** Frey & Osborne (2013)
First systematic occupation-level automation risk scores; criticized for coarse task treatment.



The task-based framework (Autor, ALM)

Every job is a bundle of tasks. Each task is classifiable along two axes:

	Cognitive	Manual
Routine	Bookkeeping, data entry, filing	Assembly line, machine operation
Non-routine	Analysis, management, writing, teaching	Home health, food prep, cleaning

- Pre-LLM: **routine** tasks are automatable (rules codifiable); **non-routine cognitive** is safe.
- This framework predicted **employment polarization**: middle collapses, top and bottom grow.
- **Eloundou et al. challenges this**: LLMs threaten *non-routine cognitive* work.

Why Frey–Osborne (2013) is not enough

- Frey–Osborne assigned each occupation a single “computerizability” probability using expert judgment.
- **Problem 1:** occupation-level, not task-level, so misses within-occupation heterogeneity.
- **Problem 2:** anchored to pre-2013 ML capabilities, so it became stale fast.
- **Problem 3:** assumes full automation is the relevant threshold, so it misses augmentation.

Eloundou et al. fixes all three

- Task-level (O*NET DWAs), not occupation-level
- Explicit LLM-capability calibration via GPT-4
- Continuous exposure score (fraction of tasks affected), not binary automation

Two meanings of GPT

Generative Pre-trained Transformer

- Transformer architecture (Vaswani et al. 2017)
- Pre-trained on large text corpora
- Fine-tuned with human feedback (RLHF)
- GPT-3.5 / GPT-4 in this paper

General Purpose Technology (Bresnahan & Trajtenberg 1995)

A technology qualifies as a GPT if it has:

- 1 **Pervasiveness:** spreads across many sectors
- 2 **Improvement:** gets better over time
- 3 **Innovation-spawning:** enables downstream innovation

Historical examples: steam engine, electricity, computer.

The paper's punning claim: the first kind of GPT is also the second kind.

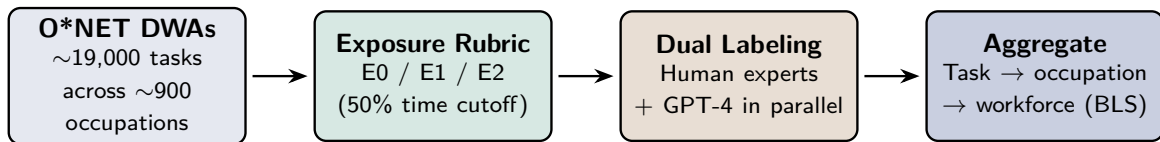
Why classification as a GPT matters economically

- GPTs generate **large, delayed macroeconomic effects**, mediated by co-invention (Bresnahan & Trajtenberg 1995; David 1990, “Computer and the Dynamo”).
- The economic productivity of electricity took ~ 30 years to materialize because factories had to be redesigned around it.
- Similarly, LLM productivity gains depend on **complementary innovations**: workflows, software wrappers, organizational restructuring.

Implication for policy & measurement

Short-run headline exposure numbers may substantially understate long-run economic transformation, because the technology's impact is enabled by complementary innovation that has barely begun.

Methodology at a glance



The design mirrors the **task-based framework literally**: use an official government task taxonomy, define a capability-relevant exposure standard, apply both a human and a machine labeler, then roll up to the workforce level.

Building block 1: O*NET Detailed Work Activities

- **O*NET** = U.S. Department of Labor's occupational information database.
- Each occupation is decomposed into **Detailed Work Activities (DWAs)**.
- Example DWAs:
 - *"Prepare tax returns for individuals or organizations"*
 - *"Read materials to determine needed actions"*
 - *"Confer with coworkers to coordinate work activities"*
- ~19,000 DWAs across ~900 occupations, so each occupation is a weighted bundle of tasks.

Why this unit choice matters

Choosing the task, not the occupation, as the unit *is* the methodological contribution. It respects Autor's warning: an occupation is not a fixed object; the tasks within it are what get automated or augmented.

Building block 2: The exposure rubric

Level	Label	Definition
E0	No exposure	LLM/LLM-based system cannot reduce task completion time by $\geq 50\%$ at equivalent quality.
E1	Direct exposure	LLM <i>alone</i> (no additional software) can reduce task time by $\geq 50\%$.
E2	Software-mediated exposure	LLM + supporting tools (retrieval, code execution, image analysis, etc.) can reduce task time by $\geq 50\%$.

- Threshold: **50% time reduction at equivalent quality**. Deliberately conservative.
- Aggregated exposure score for occupation j : fraction of DWAs labeled E1 or E2 (weighted).
- Two derived measures often quoted: α (E1-only, direct LLM), β (E1 + 50% of E2, LLM plus plausible tooling).

Building block 3: Dual labeling for validity

Human annotators

- Trained on the rubric
- Domain experts + workforce researchers
- Independent labels per DWA

GPT-4

- Given the same rubric
- Labels each DWA
- Zero-shot prompting

- **High agreement** between human and GPT-4 exposure scores at the occupation level (correlation reported strongly positive).
- Two independent labelers with different biases give some evidence of measurement validity.
- **Caveat:** GPT-4 labeling exposure to its *own* capability is subtly self-referential. High agreement is reassuring but not dispositive; humans and GPT-4 may share blind spots.

Headline results

Workforce-level exposure

- $\sim 80\%$ of U.S. workers have $\geq 10\%$ of tasks E1-exposed.
- $\sim 19\%$ of workers have $\geq 50\%$ of tasks E1-exposed.
- Direct LLM (α): accelerates $\sim 15\%$ of all worker tasks.
- LLM + software (β): 47–56% of all worker tasks.

What these numbers mean (and don't)

Do mean: the LLM *could* substantially accelerate this share of tasks.

Don't mean: this share of jobs will be destroyed, or wages compressed, or productivity gained. The paper measures **exposure** (a technological potential), not **outcome**.

Who is exposed: by wage

- Exposure **rises with wage**, reversing the historical pattern of automation.
- Prior automation waves (routinization, robots) primarily hit *middle-wage* occupations.
- LLM exposure is greatest among *high-wage cognitive* occupations.

Most-exposed occupations

Mathematicians · tax preparers · financial quantitative analysts · writers and authors · web designers · accountants · journalists · translators · public-relations specialists · legal secretaries.

Least-exposed occupations

Cooks (short-order) · dishwashers · auto glass installers · agricultural workers · slaughterers · roustabouts · painters (construction) · mostly manual + sciences requiring critical thinking.

Who is exposed: by education & industry

- Exposure **rises with education**: workers with bachelor's degrees or higher have more exposed tasks than workers without.
- Exposure **rises with barrier-to-entry occupations** that require licensing / credentials.
- Industries most exposed: *information, finance & insurance, professional & technical services*.
- Industries least exposed: *food services, agriculture, construction*.

The reversal of the polarization prediction

Autor's task framework predicted the middle would keep hollowing out. LLM exposure suggests the *top* of the wage distribution is now most vulnerable to automation-side pressure. Whether this becomes wage compression, augmentation, or something else is an open question.

What is *not* exposed

- **Manual and physical work:** hard to codify, hard to offshore, and LLMs don't have bodies. Cooks, cleaners, health aides.
- **Science-intensive work:** physicists, biologists, chemists, engineers.
The authors argue: tasks requiring *original hypothesis generation and empirical verification* remain low-exposure.
- **Critical-thinking-intensive occupations:** where the task is deep judgment under uncertainty rather than skilled synthesis.

Pattern to notice

The paper's low-exposure category is *not* the same as Autor's "non-routine manual." It's non-routine manual **plus** science-intensive cognitive work. This is a new distinction the LLM era introduces.

Do LLMs meet the classical GPT criteria?

Criterion (B&T 1995)	Historical GPT examples	LLMs (paper's claim)
Pervasiveness	Electricity used everywhere	Applicable across essentially all knowledge-work sectors (paper's exposure map is the evidence)
Improvement over time	Steam engine efficiency rose 100x	Rapid, ongoing capability gains; scaling laws
Innovation-spawning	Electricity enabled the assembly line, appliances, etc.	Enables new software categories, agentic workflows, retrieval-augmented systems

Paper's claim

LLMs plausibly qualify as GPTs. If so, the standard GPT playbook applies: **large but delayed** macro effects, mediated by **complementary innovation**.

Implication: the delayed impact

- Electricity was invented in the 1870s; U.S. productivity boom from it: 1920s.
- Computer arrived in the 1950s; the productivity boom: mid-1990s.
- Delayed by $\sim 30\text{--}40$ years in both cases, awaiting organizational reinvention.

So what does this mean for the exposure numbers?

The $\sim 80\%$ figure may substantially *understate* the long-run transformation, because it does not model the follow-on innovation that GPTs typically spawn. Alternatively, delayed impact means **the labor market won't feel the full effect for a decade or more**, giving policy time to adjust.

The paper's own caveats

- **Exposure $\neq$ automation $\neq$ displacement.** The rubric measures technological potential, not realized labor market outcomes.
- **Static capability snapshot.** Calibrated to GPT-4 era. Agentic tool use, multimodality, extended context have plausibly expanded exposure since.
- **Adoption is not modeled.** Universal-access assumption; real firms adopt slowly and unevenly.
- **Uses U.S. O*NET only.** Non-U.S. occupational mix may look different.
- **50% time-reduction threshold is a convention,** not a validated behavioral cutoff.

Additional concerns worth raising

- **Self-referential labeling.** GPT-4 rates GPT-4's capabilities. High human-model agreement is comforting but not proof of external validity.
- **No reinstatement modeled** (Acemoglu & Restrepo's key mechanism). What new tasks does the LLM era create? Not measured here.
- **Task boundaries assumed fixed.** Autor's second methodological pitfall: if LLMs reshape what tasks exist, the exposure map goes stale.
- **Occupation vs. task-cluster.** LLMs may operate at sub-task level (drafting) or super-task level (multi-step workflows). The DWA unit may not fit either well.
- **Quality preservation is assumed measurable.** What counts as “equivalent quality” for e.g. writing is contestable and context-dependent.

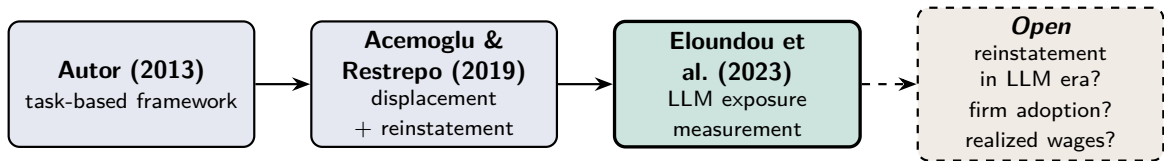
What has happened since (2023–2026)

- **Extending GPTs are GPTs to Firms** (GovAI 2024): brings the measurement to the firm-adoption level.
- **Bruegel WP 06/2024**: European labor market exposure using the same methodology.
- **Follow-on econometric work**: trying to identify whether high-exposure occupations actually saw employment or wage changes 2023–2025. Early evidence is mixed; some highly-exposed occupations (e.g., writing) have *grown*.
- **Startup-based measures** (arXiv 2412.04924): infer exposure from what AI startups actually target.
- **AI Exposure Scores critique** (arXiv 2606.23633): what these scores measure vs. miss.

The methodology has become a standard

The exposure-rubric approach is now the default in the field, but researchers are increasingly aware it captures only *potential*, not *realized* effects.

Position in the broader arc



- Eloundou et al. operationalizes Autor's framework for LLMs.
- But leaves Acemoglu & Restrepo's reinstatement dynamic entirely unmodeled.
- That gap is where the next generation of research is happening.

Prompts for discussion (1/2)

- 1 Eloundou et al. **reverses the polarization prediction**: high-wage cognitive work is now most exposed. Does this mean Autor's polarization era is ending, or does high exposure not translate into high displacement risk because high-wage workers capture productivity gains?
- 2 Is LLM adoption a paradigm case of Acemoglu & Restrepo's "**so-so automation**" concern: lots of displacement potential, unclear reinstatement, modest measured productivity gains so far?
- 3 Autor's second methodological pitfall (treating tasks as fixed) seems especially serious here. If LLMs reshape what tasks even exist, the exposure map goes stale fast. Should exposure be measured **dynamically**?

Prompts for discussion (2/2)

- 4 The paper works at the task/occupation level. Adoption is a **firm-level decision**. Is the firm the natural next frontier (GovAI follow-up)?
- 5 The rubric's threshold is **50% time reduction**. Is that the right cutoff for labor-market impact? What alternative thresholds would you defend?
- 6 If GPT-4 labeling its own capability is subtly self-referential, how would you **externally validate** an LLM exposure rubric?
- 7 Three years post-paper: some highly-exposed occupations (writing, translation) have *grown*, not shrunk. What are the possible explanations?

Pick two. Leave time for the audience to disagree.

Discussion.

Eloundou, Manning, Mishkin, Rock (2023).
arXiv:2303.10130 · Published in *Science* 2024